

Cost of Capital and Firm Value

Capital Structure and Dividend Policy



Definitions

Core concepts for financing and valuation

Capital structure

The relative proportion of debt, equity and other securities outstanding.

In this course, we focus on the mix of debt and equity.

Cost of capital

The opportunity cost of investing in the portfolio of the firm's capital structure.

Think: expected return on an alternative investment with the same risk.

Applied interpretation: financing has a cost because investors can place their money elsewhere in investments with comparable risk.

Important Questions

The questions that organize today's session

1 Firm value

Can changing the proportion of debt and equity increase firm value?

2 Cost of capital

Can capital structure reduce the WACC used to discount expected cash flows?

3 Optimal structure

If financing matters, under which conditions — and what would “optimal” mean?

VALUATION LINK

Firm value = Present value of expected cash flows

These questions will be answered progressively: perfect markets first, then taxes.

Capital Structure: Perfect Capital Markets

The benchmark assumptions

No transaction costs

Investors can buy, sell and short securities without trading frictions.

No income taxes

Financing choices do not change tax payments.

No bankruptcy costs

Financial distress does not destroy value.

FIRM VALUE

$$V = D + E$$

This is a benchmark, not a description of the real world. It isolates the pure effect of changing financial claims.

Example: Unlevered Corporation U vs Levered Corporation L

A simple perfect-capital-market example

OPERATING CASH FLOWS

Strong state: \$6,000

Weak state: \$4,000

Each state has 50% probability.

Expected operating cash flow = \$5,000

UNLEVERED U

Debt = \$0

All-equity financed

Equity receives the entire operating cash flow.

LEVERED L

Debt = \$2,000

Risk-free interest rate = 10%

Debt repayment = \$2,200

Equity is the residual claimant.

Question: Does the different financing mix change the total payoff of the firm?

Solution: Same operating payoff — different allocation

U — EQUITY PAYOFF

$$50\% \times \$6,000 + 50\% \times \$4,000$$
$$= \$5,000$$

L — DEBT PAYOFF

Debt receives \$2,200 in either state.

Expected debt payoff = \$2,200

L — EQUITY PAYOFF

Strong: $\$6,000 - \$2,200 = \$3,800$
Weak: $\$4,000 - \$2,200 = \$1,800$

Expected = \$2,800

LEVERED FIRM

\$2,200 debt + \$2,800 equity = \$5,000 total payoff

What does the example tell us?

Same total payoff

U: \$5,000

L: \$2,200 + \$2,800 = \$5,000

Same firm value

If the payoffs are identical, arbitrage implies the firms must have the same value.

Same WACC

The financing mix does not change the required return on the operating assets.

But the required return on L's equity is higher.

Why?

Debt receives a fixed claim first. Equity is the residual claim and therefore becomes riskier when leverage is introduced.

Modigliani–Miller intuition

Keep the intuition; skip the arbitrage proof

WHOLE MILK

The operating business before leverage.

CREAM

Debt: a contractual claim with priority.

SKIM MILK

Levered equity: the residual, now riskier.

MM INTUITION

Debt value + Equity value = Total firm value

Separating the claims does not create more operating cash flow.

M&M Proposition 1

PERFECT CAPITAL MARKETS

$$V_U = V_L$$

Firm value

Independent of capital structure.

WACC

Independent of capital structure.

Economic reason

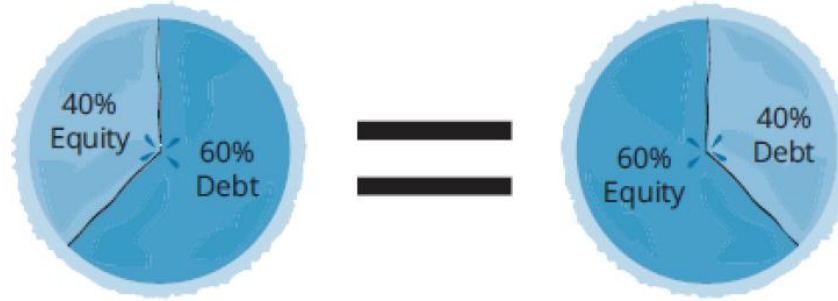
Financing does not change operating cash flows.

Does the Financing Mix Change the Pie?

Vote first: Firm A has 60% debt; Firm B has 40% debt. Same operating assets and cash flows.

YOUR PREDICTION

- A. Firm A is worth more
- B. Firm B is worth more
- C. Same firm value



THEN EXPLAIN

What changed?
What did NOT change?

Use the cash-flow logic of MM Proposition I.

Application: Debt-financed stock repurchase

Debt-financed recapitalization example

BEFORE REPURCHASE

Cash = \$0m

Existing assets = \$20,000m

Total assets = \$20,000m

Shares outstanding = 1,000m

Price per share = \$20

TRANSACTION

Issue \$5,000m of debt

Use all proceeds to
repurchase shares

QUESTIONS

What is firm value after?

What is equity value after?

How many shares are repurchased?

What happens to price per share?

Stock repurchase: Solution

FIRM VALUE

Existing assets remain \$20,000m.

Under MM Proposition 1:

$$V = \$20,000\text{m}$$

EQUITY VALUE

Debt after transaction = \$5,000m

$$\begin{aligned} E &= V - D \\ &= \$20,000\text{m} - \$5,000\text{m} \\ &= \$15,000\text{m} \end{aligned}$$

SHARES & PRICE

At \$20 per share, \$5,000m repurchases 250m shares.

Shares remaining = 750m
Price per share = \$20

The share price does not rise merely because the company replaces equity with debt.

Application: Stock issuance

Equity-financing example

BEFORE ISSUANCE

Cash = \$0m
Existing assets = \$20,000m
Total assets = \$20,000m
Shares outstanding = 1,000m
Price per share = \$20

TRANSACTION

Issue \$5,000m of new equity

Cash raised remains inside
the firm for expansion

SHAREHOLDER CONCERN

Will issuing new shares dilute
ownership and reduce the value per
existing share?

Stock issuance: Solution

CASH & ASSETS

Cash raised = \$5,000m
Existing assets = \$20,000m

Total equity value after issuance =
\$25,000m

NEW SHARES

At \$20 per share:
 $\$5,000\text{m} \div \$20 = 250\text{m}$ new shares

Total shares = 1,250m

PRICE PER SHARE

$\$25,000\text{m} \div 1,250\text{m}$ shares
= \$20 per share

Issuing fairly priced equity increases the number of shares and the assets backing them proportionally.

M&M Proposition 2

Leverage increases the required return on equity

$$r_E = r_{WACC} + (D/E) \times (r_{WACC} - r_D)$$

Business risk

Captured by the required return on the firm's operating assets.

Financial leverage

Higher D/E increases the risk borne by equity.

Equity return

Shareholders demand a higher expected return as leverage rises.

What Happens as Leverage Increases?

Perfect capital markets, no corporate taxes. Predict the three lines before we interpret the graph.

PREDICT

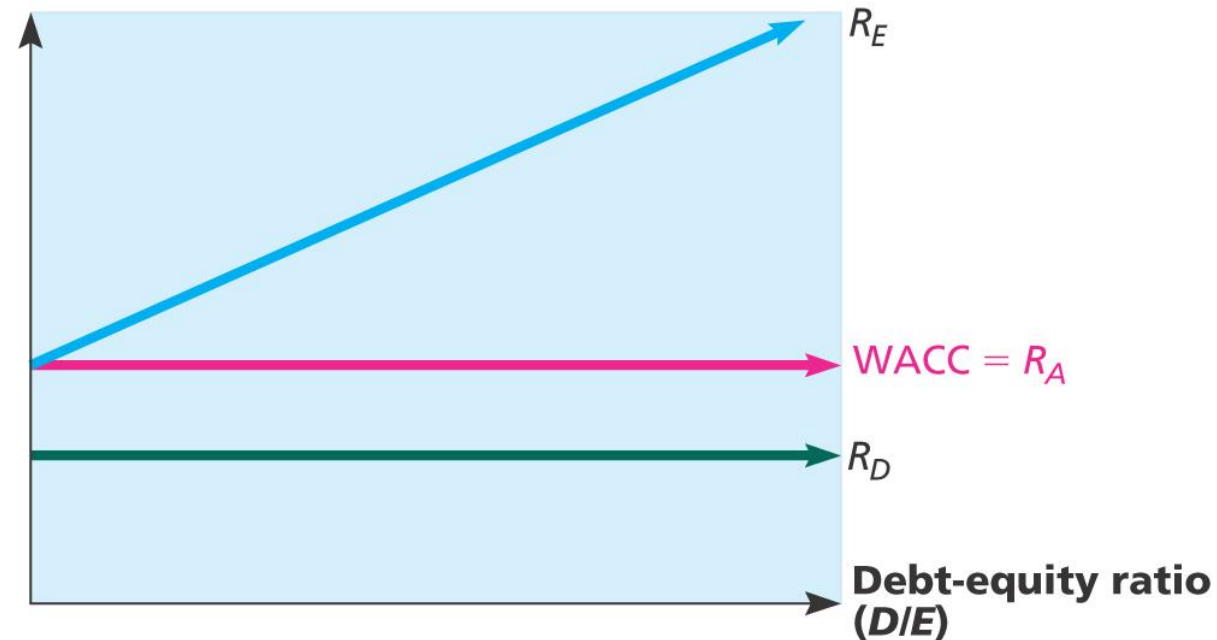
As D/E rises:

r_D ?

r_E ?

WACC ?

Cost of capital
(%)



WHY?

If debt is cheaper,
why does WACC not
fall?

$R_E = R_A + (R_A - R_D) \times (D/E)$ by M&M Proposition II

$$R_A = \text{WACC} = \left(\frac{E}{V}\right) \times R_E + \left(\frac{D}{V}\right) \times R_D$$

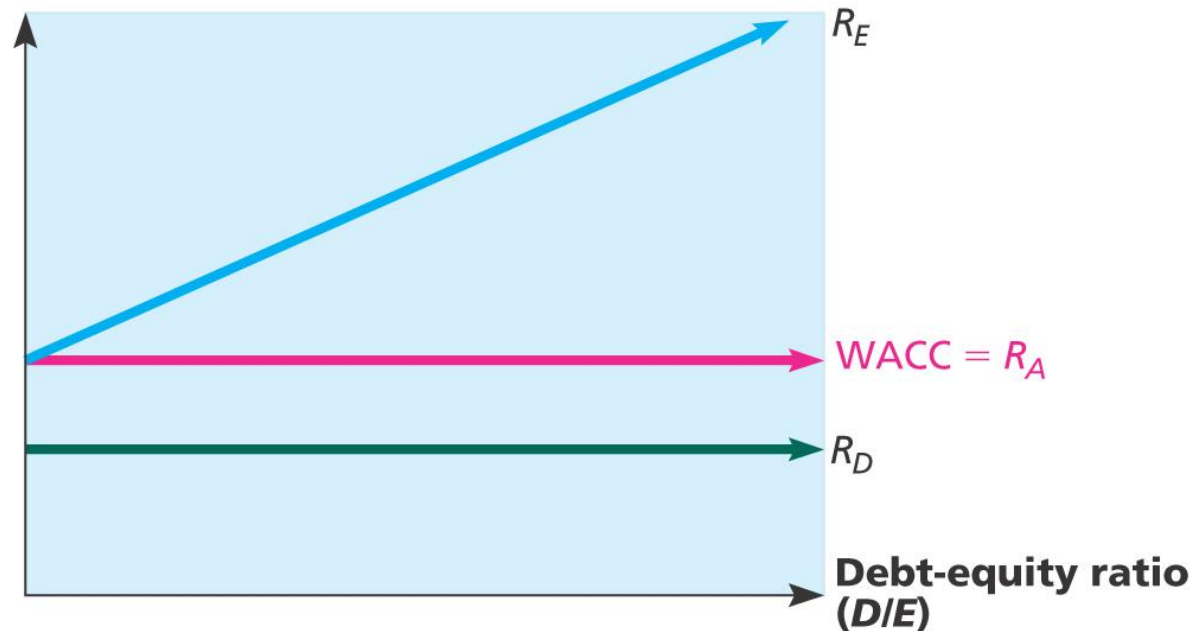
where $V = D + E$

Discuss for 60 seconds with the person next to you.

Graph Interpretation — MM Proposition II

The operating business is unchanged; the risk is redistributed toward equity.

Cost of capital (%)



$$R_E = R_A + (R_A - R_D) \times (D/E) \text{ by M\&M Proposition II}$$

$$R_A = \text{WACC} = \left(\frac{E}{V}\right) \times R_E + \left(\frac{D}{V}\right) \times R_D$$

where $V = D + E$

1 COST OF DEBT

r_D stays constant in this benchmark.

2 COST OF EQUITY

r_E rises with D/E because equity becomes riskier.

3 WACC

WACC remains constant: financing does not change asset risk.

Cheaper debt $\neq$ cheaper firm financing.

Perfect-market example revisited

Use MM Proposition 2

GIVEN

Expected cash flow = \$5,000

WACC = 100%

Debt in L = \$2,000

Risk-free debt rate = 10%

U is all equity; L is levered.

VALUES

$V_U = V_L = \$2,500$

Debt value = \$2,000

Equity value = \$500

EQUITY RETURN

Expected equity payoff = \$2,800

$r_E = \$2,800 / \$500 - 1$
 $= 460\%$

Leverage leaves total value unchanged but makes the remaining equity dramatically riskier.

Capital Budgeting and WACC

Debt-financed recapitalization

CURRENT

$D/E = 0.6$

Risk-free debt rate = 12%

Current required return on equity = 20%

No corporate taxes in this part.

TRANSACTION

Debt-financed stock
repurchase

Target $D/E = 1.0$

QUESTIONS

1. Current WACC?
2. New required return on equity?
3. New WACC?

Capital Budgeting and WACC: Solution

Perfect capital markets — no taxes

CURRENT WACC

$$\begin{aligned} D/E = 0.6 &\Rightarrow D/V = 0.6/1.6 = 37.5\% \\ E/V &= 62.5\% \end{aligned}$$

$$\begin{aligned} WACC &= 37.5\% \times 12\% + 62.5\% \times 20\% \\ &= 17\% \end{aligned}$$

NEW EQUITY RETURN

At $D/E = 1$:

$$\begin{aligned} r_E &= 17\% + 1 \times (17\% - 12\%) \\ &= 22\% \end{aligned}$$

NEW WACC

$$\begin{aligned} D/V &= 50\% \\ E/V &= 50\% \end{aligned}$$

$$\begin{aligned} WACC &= 50\% \times 12\% + 50\% \times 22\% \\ &= 17\% \end{aligned}$$

More debt → higher equity return → WACC stays at 17%.

Levered and Unlevered Betas

Expressing the same leverage effect through risk

NO-TAX RELATIONSHIP

$$\beta_A = \beta_E / (1 + D/E) \quad | \quad \beta_E = \beta_A \times (1 + D/E)$$

Asset beta β_A

Risk of the operating assets — before financing leverage.

Equity beta β_E

Risk borne by shareholders after leverage.

Leverage effect

Higher D/E raises equity beta even when asset risk is unchanged.

Beta application: M&M Corporation

Leverage example revisited

CURRENT EQUITY BETA

CAPM inputs:

$r_E = 20\%$

$r_F = 12\%$

Market risk premium = 5%

$$\beta_E = (20\% - 12\%) / 5\% = 1.6$$

ASSET BETA

Current $D/E = 0.6$

$$\begin{aligned}\beta_A &= 1.6 / (1 + 0.6) \\ &= 1.0\end{aligned}$$

AFTER $D/E = 1$

$$\begin{aligned}\beta_E &= 1.0 \times (1 + 1) \\ &= 2.0\end{aligned}$$

$$\begin{aligned}\text{New } r_E &= 12\% + 2 \times 5\% \\ &= 22\%\end{aligned}$$

Same conclusion as MM Proposition 2: leverage raises equity risk and required return.

Capital Structure with Taxes

Now relax one perfect-market assumption

Interest is deductible

Corporate profits are taxed after deductible interest payments.

Taxes fall

Interest expense reduces the amount of corporate tax paid.

Cash to investors rises

Lower taxes leave more cash flow available to the firm's owners.

This creates an incentive to use debt.

Key change from MM without taxes

Financing now changes total cash flow because it changes the amount paid to the government.

Modified M&M Proposition 1

Value of the interest tax shield

PERMANENT DEBT BENCHMARK

$$VL = VU + \tau D$$

VU

Value of the unlevered firm.

τD

Present value of the interest tax shield under the lecture-note assumptions.

VL

Value of the levered firm.

Debt can now increase firm value because it reduces taxes.

Interactive Check — Corporate Taxes Change What?

Now interest is tax deductible. Before seeing the graph, predict the effect of more debt on firm value.

STEP 1

More debt
→ more interest expense
→ larger tax deduction

STEP 2

Tax payments
increase / decrease / stay the
same?

STEP 3

Firm value
increase / decrease / stay the
same?

Question: What is different from the no-tax MM world?

Hint: financing now changes a real cash flow — the amount paid to the government.

Interest tax shield — a simple cash-flow view

UNLEVERED

Operating cash flow before tax = X

Taxable amount = X

After-tax cash flow:
 $CU = X(1 - \tau)$

LEVERED

Interest = rD

Interest reduces taxable income.

Investors receive operating cash after tax plus debt interest.

EXTRA CASH

Annual tax saving:

$\tau \times rD$

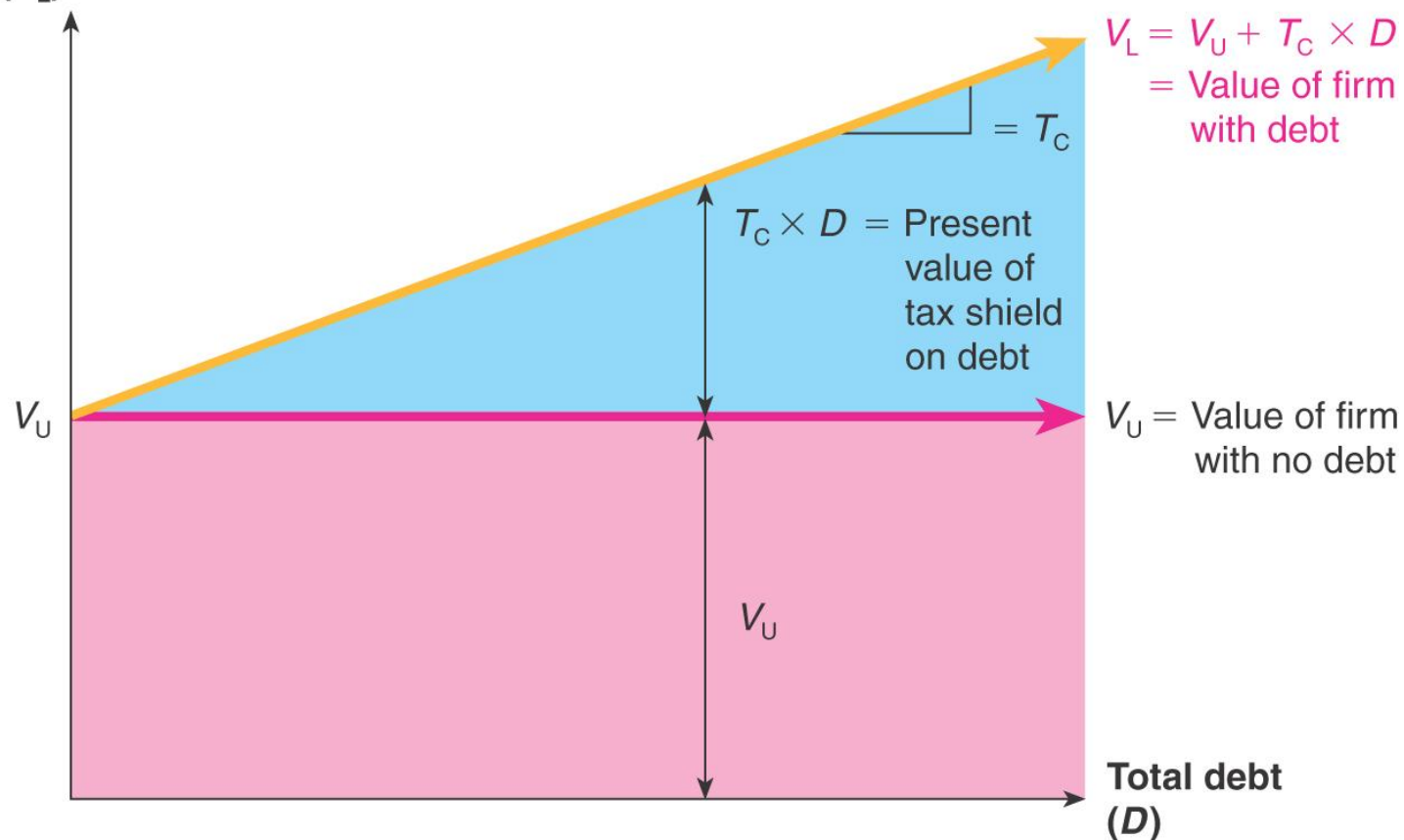
= Interest tax shield

The tax shield is a cash-flow benefit — not merely a lower accounting tax expense.

Graph — Debt Creates Value Through the Tax Shield

Under the permanent-debt tax benchmark: $V_L = V_U + T_C \times D$

Value of
the firm
(V_L)



STARTING POINT

At $D = 0$, $V_L = V_U$.

SLOPE

Each additional €1 of permanent debt adds T_C of value.

INTERPRETATION

The vertical gap is the present value of the interest tax shield.

What assumption makes this line keep rising?

The value of the firm increases as total debt increases because of the interest tax shield. This is the basis of M&M Proposition I with taxes.

Modified M&M Proposition 2

Leverage and equity return with corporate taxes

$$r_E = r_U + (D/E) \times (r_U - r_D) \times (1 - \tau)$$

Leverage still matters

More debt still increases the financial risk of equity.

Tax shield moderates the effect

The leverage premium is multiplied by $(1 - \tau)$.

After-tax WACC

Can fall as debt creates tax benefits.

After-tax WACC

After-tax financing cost

AFTER CORPORATE TAX

$$\text{WACC} = (D/V) \times r_D \times (1 - \tau) + (E/V) \times r_E$$

Debt weight

Use D/V .

After-tax debt cost

$r_D \times (1 - \tau)$

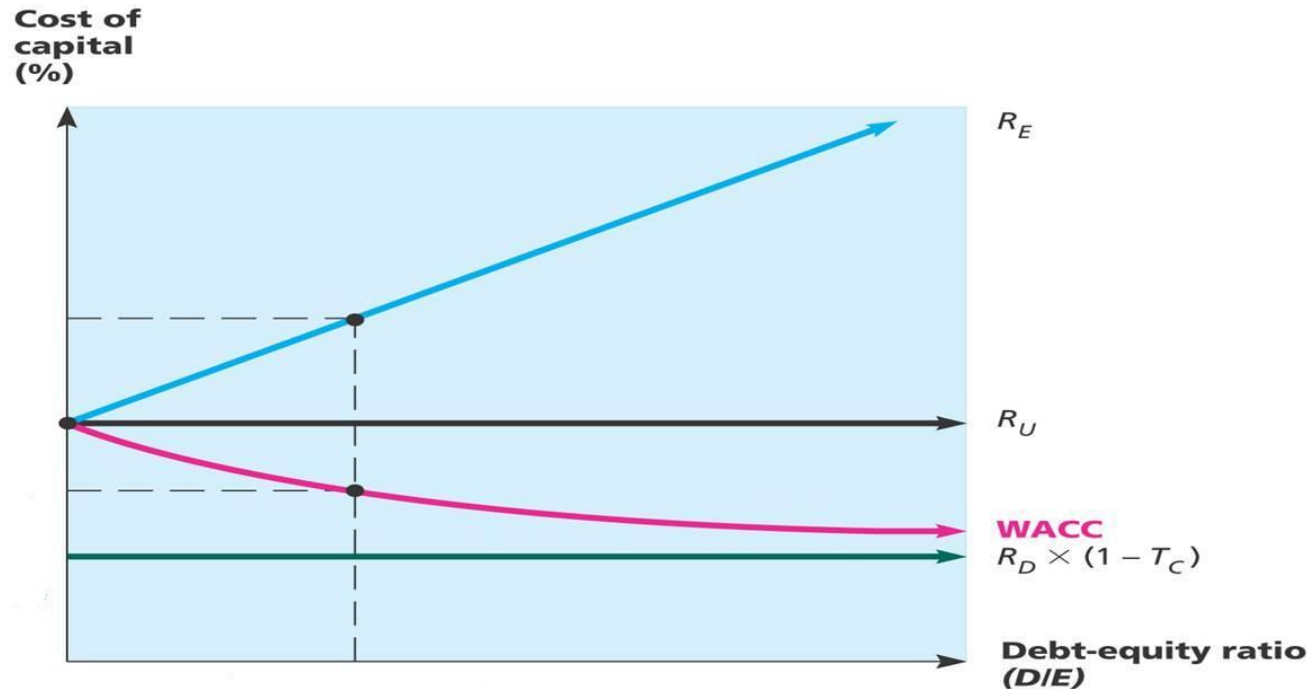
Equity weight

Use E/V and the levered required return on equity.

Applied warning: when leverage changes, update both the weights and the required return on equity.

Interactive Graph — WACC with Corporate Taxes

Predict first: Which line rises, which falls, and which stays constant?



M&M Proposition I with taxes implies that a firm's WACC decreases as the firm relies more heavily on debt financing:

$$WACC = \left(\frac{E}{V}\right) \times R_E + \left(\frac{D}{V}\right) \times R_D \times (1 - T_C)$$

M&M Proposition II with taxes implies that a firm's cost of equity R_E rises as the firm relies more heavily on debt financing:

$$R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C)$$

r_E

Rises as leverage increases.

$r_D(1 - T_C)$

Constant in this benchmark.

WACC

Falls because debt provides a tax advantage.

Compare this with the no-tax graph.

Capital Budgeting and WACC with Taxes

Recapitalization with corporate taxes

CURRENT INPUTS

$$D/E = 0.6$$

$$rD = 12\%$$

$$rE = 20\%$$

Corporate tax rate = 40%

Target after repurchase:

$$D/E = 1.0$$

CURRENT WACC

$$D/V = 37.5\%$$

$$E/V = 62.5\%$$

$$\begin{aligned} \text{WACC} &= 37.5\% \times 12\% \times (1 - 40\%) \\ &+ 62.5\% \times 20\% \end{aligned}$$

$$= 15.20\%$$

WHAT NEXT?

To evaluate the new structure we need:

1. Unlevered required return
2. New levered equity return
3. New after-tax WACC

Capital Budgeting and WACC with Taxes: Solution

1 UNLEVERED RETURN

From the current financing structure:

$$r_U \approx 17.8824\%$$

2 NEW EQUITY RETURN

At $D/E = 1$:

$$\begin{aligned} r_E &= 17.8824\% + \\ &1 \times (17.8824\% - 12\%) \times (1 - 40\%) \\ &\approx 21.41\% \end{aligned}$$

3 NEW WACC

$$D/V = E/V = 50\%$$

$$\begin{aligned} \text{WACC} &= 50\% \times 12\% \times (1 - 40\%) + \\ &50\% \times 21.41\% \\ &\approx 14.305\% \end{aligned}$$

With corporate taxes: more debt can reduce after-tax WACC because of the interest tax shield.

$$r_E = r_U + \frac{D}{E}(r_U - r_D)(1 - \tau)$$

Levered and Unlevered Betas with Taxes

Beta application

UNLEVERING WITH TAXES

$$\beta_A = \beta_E / [1 + (D/E)(1 - \tau)] \mid \beta_E = \beta_A \times [1 + (D/E)(1 - \tau)]$$

CURRENT

$$\beta_E = 1.6$$

$$D/E = 0.6$$

$$\tau = 40\%$$

$$\beta_A = 1.17647$$

TARGET D/E = 1

$$\beta_E = 1.17647 \times [1 + 1 \times (1 - 40\%)]$$

$$= 1.88235$$

RETURN & WACC

$$r_E = 12\% + 1.88235 \times 5\% \\ = 21.41\%$$

$$WACC = 14.305\%$$

What Is Missing from the Tax-Only Model?

Case II gives debt a tax benefit. If that were the whole story, more debt would always look better.

CASE I — NO TAXES

Firm value unchanged.
WACC unchanged.
Debt has no net value effect.

CASE II — TAXES

Interest tax shield adds value.
Firm value rises.
WACC falls.

WHAT IS MISSING?

Why don't firms use almost 100% debt?

Debt must also create costs.

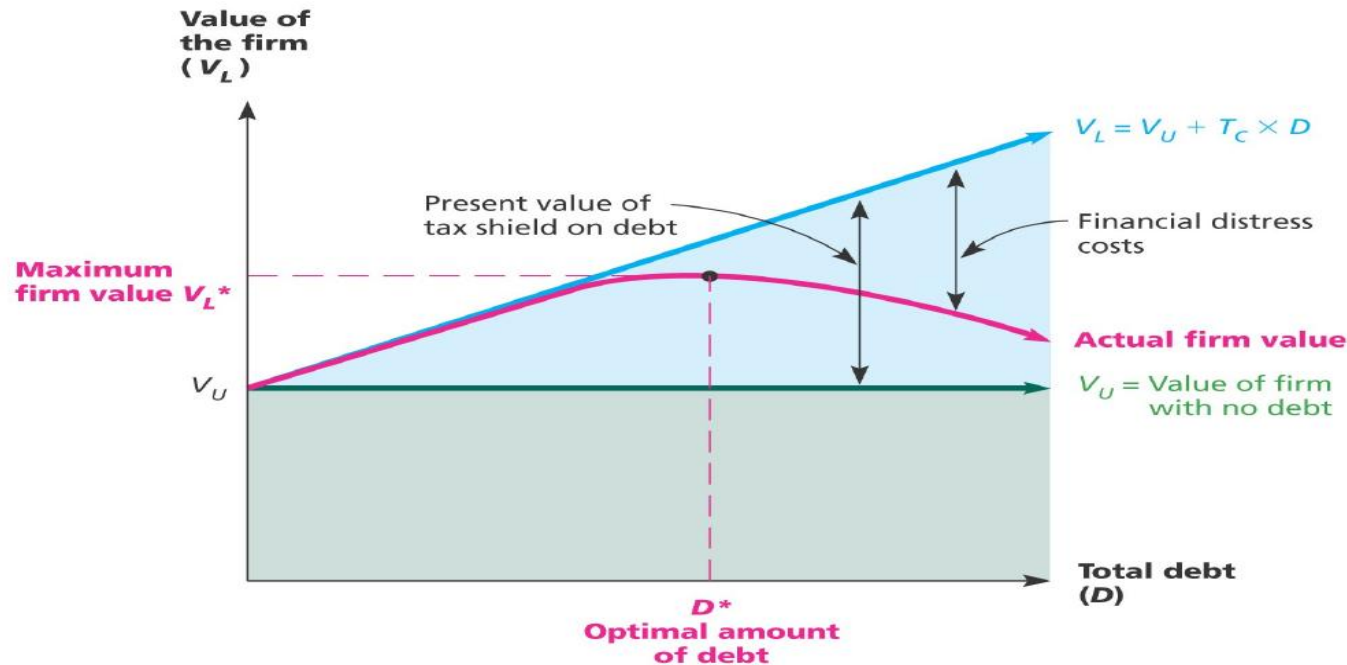
Next: add expected financial distress and bankruptcy costs.

Tax benefit of debt + financial distress cost → an optimal capital structure can emerge.

Case III — Proposition I: Firm Value with Financial Distress

Use the original Case III graph: tax benefits increase value, but distress costs eventually dominate.

CASE III – PROPOSITION I



According to the static theory, the gain from the tax shield on debt is offset by financial distress costs. An optimal capital structure exists that just balances the additional gain from leverage against the added financial distress cost.

LOW DEBT

Tax-shield benefit dominates → value rises.

D^* — OPTIMUM

Firm value reaches its maximum.

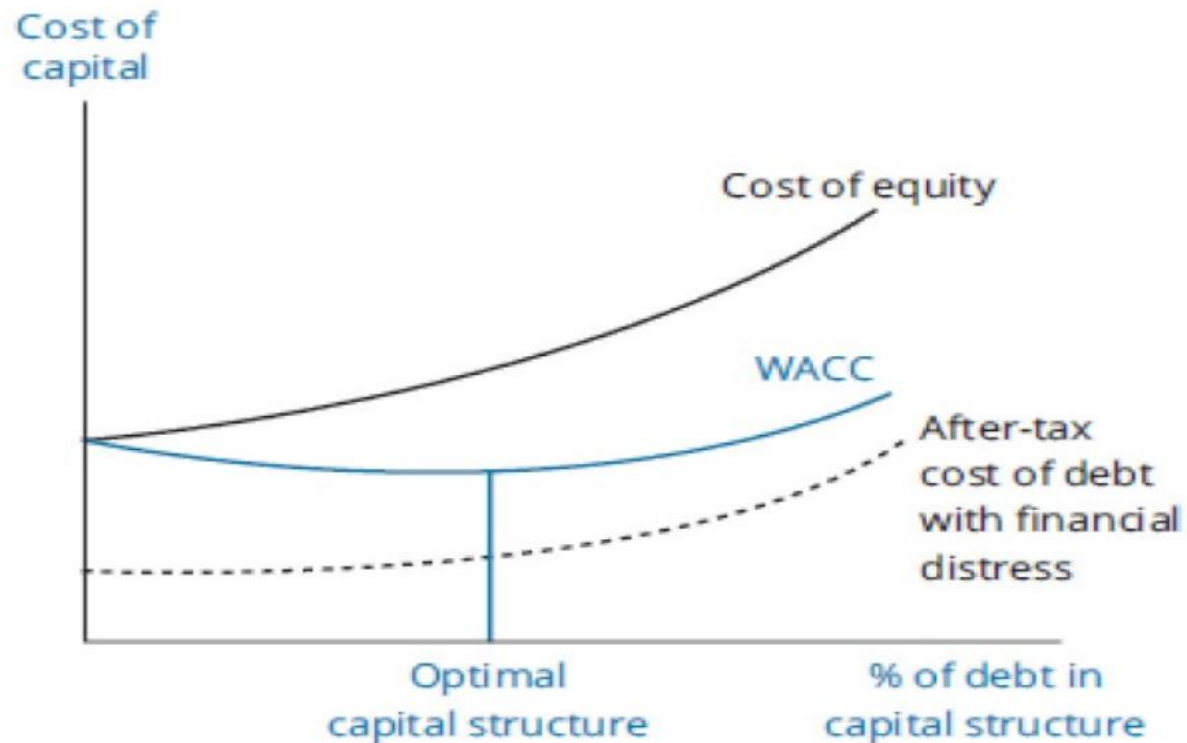
HIGH DEBT

Expected distress costs dominate → value falls.

Case III — Proposition II: WACC and Optimal Capital Structure

The same trade-off appears through the cost of capital.

CASE III — PROPOSITION II



AT FIRST

Tax advantage pushes WACC downward.

AT THE OPTIMUM

WACC reaches its minimum.

TOO MUCH DEBT

Distress risk raises financing costs and WACC.

Cases I–III — Systematic Capital-Structure Logic

Each case relaxes one assumption and changes the conclusion.

CASE I — NO TAXES

Benefit of debt: none
Cost added: none

Firm value: constant
WACC: constant

Capital structure is irrelevant.

CASE II — TAXES

Benefit of debt: tax shield
Cost added: none yet

Firm value: rises
WACC: falls

Model favors more debt.

CASE III — DISTRESS

Benefit of debt: tax shield
Cost added: expected distress

Firm value: maximum at D^*
WACC: minimum at D^*

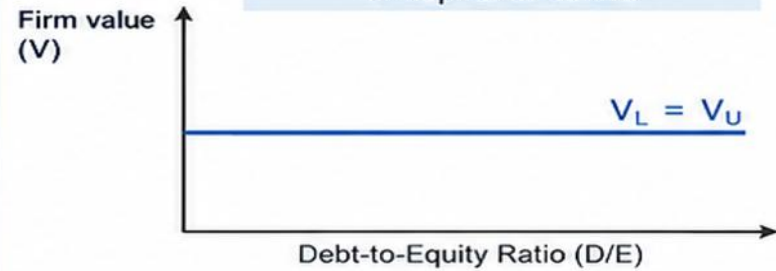
Optimal capital structure.

No effect → tax benefit → tax benefit versus distress cost → optimum

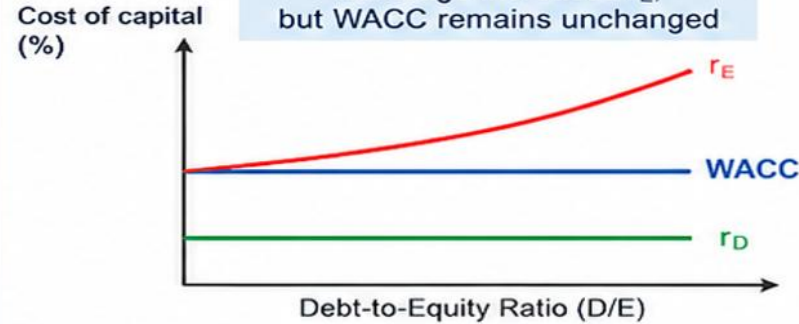
Summary: MM Propositions Across the Three Cases

Case I: No Taxes (Perfect Capital Markets)

MM Proposition I
Firm value is independent
of capital structure

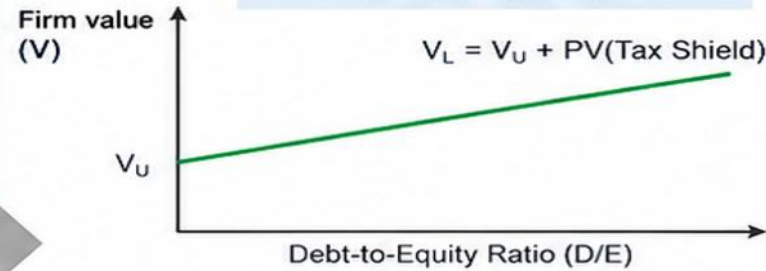


MM Proposition II
Leverage increases r_E ,
but WACC remains unchanged

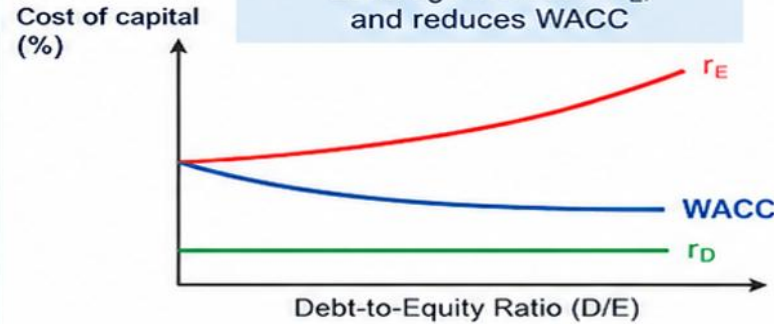


Case II: Corporate Taxes (Interest Tax Shield)

MM Proposition I
Firm value increases with leverage
due to the tax shield

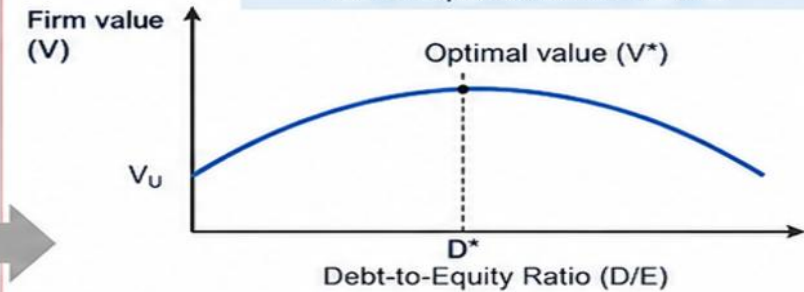


MM Proposition II
Leverage increases r_E ,
and reduces WACC

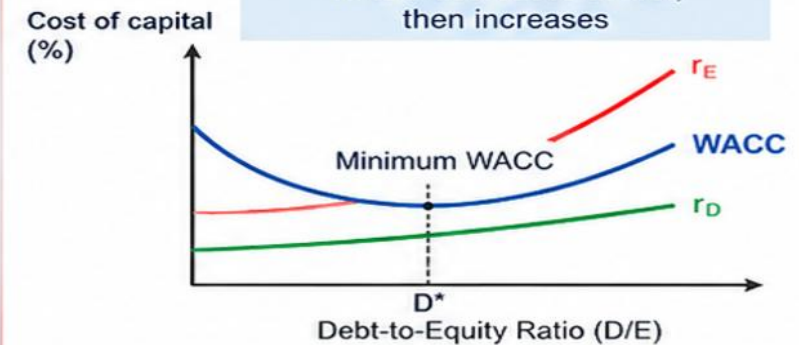


Case III: Corporate Taxes and Financial Distress Costs

MM Proposition I
Firm value increases at first, then falls
due to expected distress costs



MM Proposition II
WACC decreases at first,
then increases



Case	MM Proposition I (Firm Value)	MM Proposition II (Cost of Capital)	Key Intuition
I. No taxes	$V_L = V_U$ (independent of leverage)	r_E increases, r_D constant, WACC constant	Financing mix just reallocates value between debt and equity.
II. Corporate taxes	$V_L = V_U + PV(\text{Tax Shield})$ (increases with leverage)	r_E increases, r_D constant, WACC decreases	Interest is tax deductible, creating value through the tax shield.
III. Taxes and financial distress	V_L increases, then decreases (optimal value at D^*)	WACC decreases, then increases (minimum at D^*)	Tax benefits are offset by expected financial distress costs.